All 24 D-Flow choices are frozen

The study is unfinished. All eight Task 2 D-Flow conditions have completed literal top-three fresh40 confirmation for each independently trained model. QGF and FK each have three frozen choices. IPG remains unqualified, so the required all 33 barrier is incomplete and no new main heldout-100 test has started.

Own-best condition	lambda	Seed 11	Seed 23	Seed 37
Regular / raw Q	0	35/40	40/40	37/40
Regular / raw Q	1	28/40	40/40	40/40
Stochastic / raw Q	0	40/40	40/40	39/40
Stochastic / raw Q	1	37/40	40/40	40/40
Regular / averaged Q	0	38/40	40/40	37/40
Regular / averaged Q	1	25/40	40/40	40/40
Stochastic / averaged Q	0	40/40	40/40	40/40
Stochastic / averaged Q	1	35/40	40/40	40/40
Original QGF	-	40/40	40/40	40/40
Original FK	-	40/40	40/40	40/40
Original IPG	-	unqualified	unqualified	unqualified

Interpretation: these confirmation results do not show tuned D-Flow outperforming tuned QGF. They are independently selected settings, not matched-parameter causal effects of Q averaging and not heldout performance. The three training seeds are independent models; paired rollout starts are paired observations, not more training seeds.

QGF selected (M,kappa): seed 11 (64,16), seed 23 (8,320), seed 37 (16,16). FK selected kappa80,N32,noise .02 with M64/32/16. Equal numeric kappa across methods does not imply equal accumulated guidance.

Original DF24 aggregate marker:19d8af37; independent terminal audit:39c0d55d. The freeze queried no model, sampler or environment. Its24 choices retain all72 literal alternatives. No hidden fourth candidate was substituted.

Every selected D-Flow tuple

All rows use identity surrogate guidance, full-F8 endpoints and T=1. I is latent updates, K is Q neighbors. The final column retains the original literal three fresh40 outcomes; the selected score can be from a later-ranked candidate.

Seed	Mode/Q	L	kappa	eta	I	sigma	K	CONF	Literal3 /40
11	reg/raw	0	0.1	0.1	256	-	-	35/40	35/31/21
11	reg/raw	1	40	0.03	256	-	-	28/40	24/28/0
11	sto/raw	0	5	0.1	128	-	-	40/40	40/24/22
11	sto/raw	1	5	0.1	128	-	-	37/40	37/23/10
11	reg/avg	0	1	0.1	256	0.003	16	38/40	36/38/21
11	reg/avg	1	64	0.03	256	0.5	4	25/40	24/25/24
11	sto/avg	0	5	0.1	128	0.003	16	40/40	40/29/24
11	sto/avg	1	5	0.03	128	0.003	16	35/40	35/26/26
23	reg/raw	0	5	0.1	256	-	-	40/40	40/40/40
23	reg/raw	1	64	0.03	256	-	-	40/40	40/40/40
23	sto/raw	0	80	0.03	256	-	-	40/40	40/40/22
23	sto/raw	1	80	0.01	128	-	-	40/40	40/33/30
23	reg/avg	0	16	0.1	256	0.3	4	40/40	40/40/40
23	reg/avg	1	80	0.03	128	0.5	16	40/40	40/40/40
23	sto/avg	0	64	0.1	128	0.5	4	40/40	40/40/40
23	sto/avg	1	80	0.1	256	0.01	4	40/40	40/40/40
37	reg/raw	0	0.1	0.1	256	-	-	37/40	37/34/25
37	reg/raw	1	5	0.1	128	-	-	40/40	40/22/0
37	sto/raw	0	5	0.1	128	-	-	39/40	39/39/32
37	sto/raw	1	5	0.1	128	-	-	40/40	40/24/4
37	reg/avg	0	0.1	0.1	256	0.003	4	37/40	37/36/23
37	reg/avg	1	5	0.03	256	0.003	4	40/40	40/26/24
37	sto/avg	0	3	0.1	256	0.003	4	40/40	40/34/30
37	sto/avg	1	5	0.03	128	0.003	4	40/40	40/25/25

Full machine-readable tuples, configuration IDs, model identities, source paths and alternatives are in RESULTS.json and DF24_SELECTED_TUPLES.csv. All 27,720 TRAIN scores are preserved separately; they are not native success episodes.

DF ranges: 20 kappas from 0 to 80; eta .001/.003/.005/.01/.03/.1; I1/2/4/8/16/32/64/128/256; averaged-Q sigma .003/.01/.03/.06/.1/.2/.3/.5 and K4/16. Hierarchical 9,240 settings/model, not 73,440 Cartesian. Historical I512 and FD32 failures remain failures.

Harder Cube: trained, not evaluated

The local CubeTripleTask3 pilot uses 46-dimensional observations and a 25-dimensional action chunk: five physical controls of five coordinates. It uses the local 1M-transition data slice, unaugmented seed 11 models, 128x128 networks, H5, F8, T1. It is a limited-data adaptation, not the paper 100M-data replication.

Completed stage	Actual result	Outer parent wall
TRAIN QA	1M tuples; 996k full-H5 eligible;125 checks PASS	5.0965s
Fresh behavior cloning	50,000 BC updates; final actor parent 91a8	236.6263s
Fresh goal-specific IQL	50,000 Q/V updates; full actor state + RNG unchanged; final 2935	260.3903s
E1 callback interface	10 B20 callbacks; 69 predicates PASS; model fully unchanged	7.2159s
H5 adapter V2	FAILED before external controls: cumulative snapshot limit	4.3053s
H5 metadata reproduction	Expected FAILED; mesh_normal crosses cumulative 32 MB	4.7697s

E1 restored the actual final IQL50000 full state using the original phase validator and Store. Three full proofs and two canonical blobs matched the parent; the full actor TrainState and RNG matched BC parent 91a8. The 10 callbacks include four velocities, raw targetQ, raw value/gradient, two same-source F8 maps and K4/K16 averaging value/gradient. These are 200 padded views of 16 fixed TRAIN anchors plus 4 inactive rows, not new training or evaluation data.

What remains before a policy comparison

The sampler engineering protocol and H5 state-comparison representation are SOURCE-only work. Original build/reset/control/queue behavior must remain unchanged. All model fields must remain accounted for; the 32 MB materialized-state bound,256 MB output cap and4 GB worker limit stay in force. No native performance or eight-condition Cube sweep has run.

The metadata reproduction records one mesh_normal array of 4,685,340 bytes, taking cumulative logical size from 30,627,768 to 35,313,108 bytes. Its per-field bound passed; cumulative 32 MB failed. This locates the reproduced engineering limit, not a policy failure. The original failed run did not persist its offending field.

TRAIN rewards are per-cube goal count minus 3: 927,047 rows at -3; 71,460 at -2; 1,493 at -1; 0 joint-goal rows. This partial-goal signal is descriptive support, not proof of competence, impossibility or true Q accuracy. The pinned-Box controlled reset is not established as the original fresh-Box reset graph.

Hard cases and unresolved failures

Known-Q dimensions 8, 32 and 128

These are synthetic flow/critic fixtures with narrow-wrong, broad-wrong, legitimately narrow-good and correct-broad controls. They are not measured IQL errors or robotic returns.

All 596 finite engineering configurations executed:585 PASS / 11 FAIL. All11 failures are strong-guidance original QGF formula-agreement failures. Finite/replay/source/bounds checks passed. The complete gate remains unqualified; these are not measured robot task failures or evidence that D-Flow repairs unseen-Q errors.

Case 387, dimension 32, legitimately narrow good region,QGF M32 / kappa 80: final action max difference 1.86511415, 625/640 coordinates. Kappa0 control 386 passed with max 1.19e-7. A compact 8 B20 diagnostic reproduced all old public outputs byte-for-byte before capturing named intermediate exports. It added no independent or native trials.

SavedJSON shows 17/20 initial gradient-byte mismatches in both cases; three guided rows first differ in drift despite matching named velocity and gradient. Later guided trajectories already have different inputs. Byte flags provide no numerical magnitudes and named exports are not actual consumer-memory observations. A unique compiler/hardware cause or repair is not established.

Original IPG remains unqualified

The latest independent materialized-reference B144 gate retains 134 PASS / 10 FAIL, including 10 particle-close failures and one overlapping action-close failure. The old 33 residual failures,15 missing learned cases, dtype/lowering failures and both ridge domains remain preserved. Successful residual/source/noise/dtype checks do not waive trajectory checks. No C/native IPG or all 33 heldout release is authorized.

Task 5 QGF and unseen-action probes

Task 5 selected QGF	kappa	M	Fresh CONF
11	96	16	40/40
23	2	64	36/40
37	80	32	40/40

Task 5 had four seed 23 confirmation failures, not generic QGF collapse. Expanded DF engineering retains four latent-check failures and an inconclusive capture; no expanded native DF comparison is admitted. Standard full-route withheld-action probes had zero joint exposure; custom right-start probes did expose the region but used a custom reset and did not establish true Q error.

What Q averaging can and cannot do

One flow endpoint a can be queried at K nearby actions. The flow does not need K independent endpoints:

Qbar(s,a) = (1/K) sum_j Q(s, clip(a + sigma*epsilon_j))

The finite antithetic bank is fixed throughout each action generation. Sigma sets perturbation width; K sets the number of critic neighbors. K4 and K16 banks need not be nested. This reuses the frozen critic and adds no new transitions or observed rewards. Stochastic latent noise is a separate operation.

Averaging can soften a narrow false peak if nearby predictions are more reasonable. It can leave a broad wrong region wrong, or blur a genuinely narrow good action. It does not guarantee unseen-action Q repair. Lambda 1 means latent prior pull; it is not an L1 norm or support detector. Identity guidance is a surrogate, not a generic exact Boltzmann sampler.

Closed counts and resource accounting

Closed quantity	Amount
New native VAL/CONF episodes	30,360
External active action rows	16,185,534
New main heldout episodes	0
Measured completed outer-parent wall	45157.925126753s, once
Separate unknown preadmission charge	600 s conservative allowance; not claimed actual wall
Active caps / remaining of 86,400s	0 / 40642.074873247s

The first compact admission failed before AUTH/worker because an ACK flag was missing. Its setup wall/PID/PGID are unknown, not zero. A separately reviewed administrative supersession preserved that failure before the completed first scientific capture. Parent, unit, watchdog and sampler clocks are never added together.

Unfinished scope and delivery

Remaining: original IPG qualification and three selections; all 33 barrier; new main 100; 540 same-kappa points targeting 100 episodes per point in declared 20 increments; remaining Task 5 and Cube work. No silent cuts, heldout-driven tuning or forced winner. The initial 24 worker-hours are a reviewed tranche, not a whole-study completion promise.

This fixed plan 307 snapshot predates later source changes or numerical work. The private GitHub plan 179 handoff remains immutable and does not contain these later results. At 15 UTC Nov8 if still unfinished, the scheduled checkpoint must remain explicitly partial. Old reports, PDFs, datasets, checkpoints and archives remain unchanged.

Evidence: copied JSON/CSV/source primaries and SHA manifests accompany this report. They establish saved record provenance; no numerical payload replay was performed by the report writer. All report tables are metadata-derived.